

Data Storytelling

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Spring 2026

Week 12

Announcements

- Project Proposals are graded! Grades and feedback will be submitted by the end of the day/early tomorrow.
- Homework 2 is due next week. Don't wait until the last minute to start it! It will require you to do some data cleaning and wrangling, which can be time-consuming.

Making Maps with R

Why Make Maps?

- We've covered how to show how data can vary over time, but what about space?
- Maps help us understand spacial relationships and patterns that might not be obvious in a tabular format.
- Example: What does the geography of the SoH tell us about the current conflict between the US/Israel and Iran?

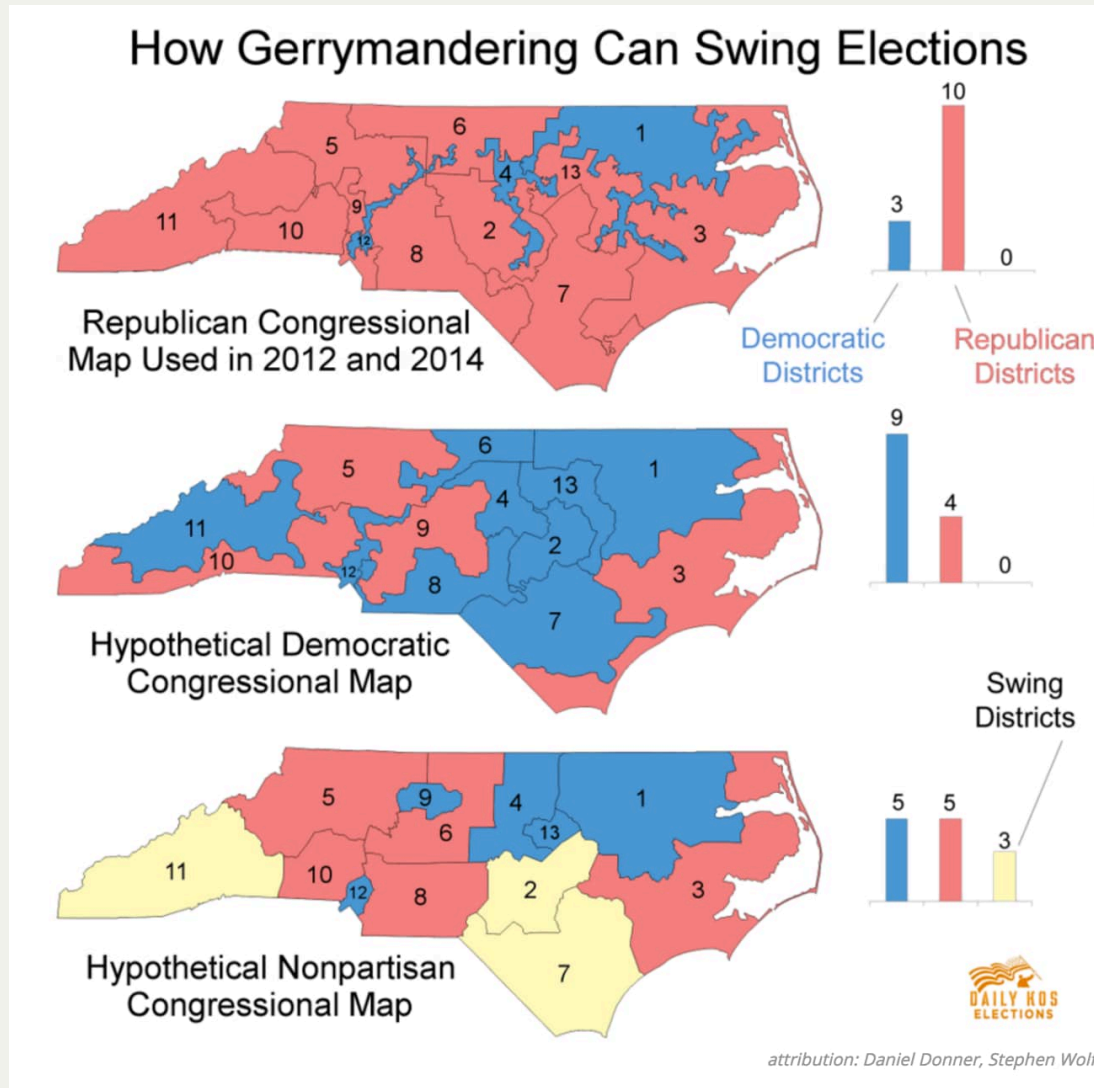
Strait of Hormuz



Power of Geography

- Geospatial data is crucial for many fields, including:
 - *Urban planning* (e.g. analyzing traffic patterns or land use)
 - *Environmental science* (e.g. tracking deforestation or climate change)
 - *Public health* (e.g. tracking disease outbreaks)
 - *Political science* (e.g. gerrymandering and geopolitics)
 - *Geology* (e.g. mapping distribution of natural resources or tectonic activity)

Examples



- Gerrymandering is a tool by which elected representatives can reshape the composition of districts in a way that maximizes their likelihood of winning
- Two techniques:
 - *Cracking*
 - Breaking groups into diluted smaller groups
 - *Packing*
 - Creating large groups of like-minded voters so that their votes are “wasted”

Projections and Coordinate Systems

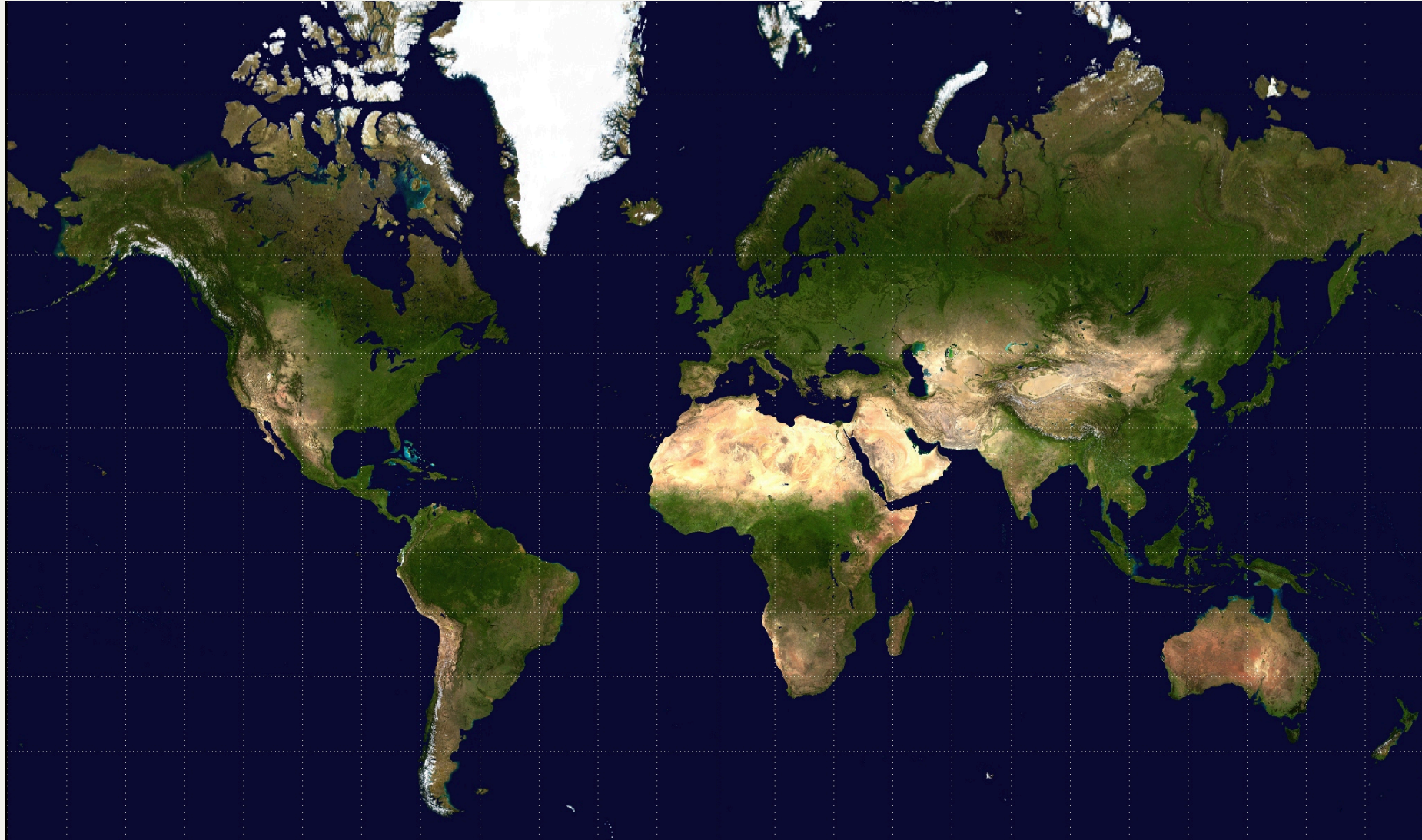
- **Coordinate Systems:** A coordinate system is a reference system that allows us to specify locations on the Earth's surface. Common coordinate systems include:
 - Geographic Coordinate System (GCS): Uses latitude and longitude to specify locations on a spherical Earth.
 - Projected Coordinate System (PCS): Uses a flat, two-dimensional surface to represent the Earth, which can lead to distortions in area, shape, distance, or direction.
- **Projections:** A mathematical transformation that converts the three-dimensional Earth into a two-dimensional map.
 - Different projections are designed to preserve different properties (e.g. area, shape, distance, direction)
 - No projection is perfectly accurate

Map Projections

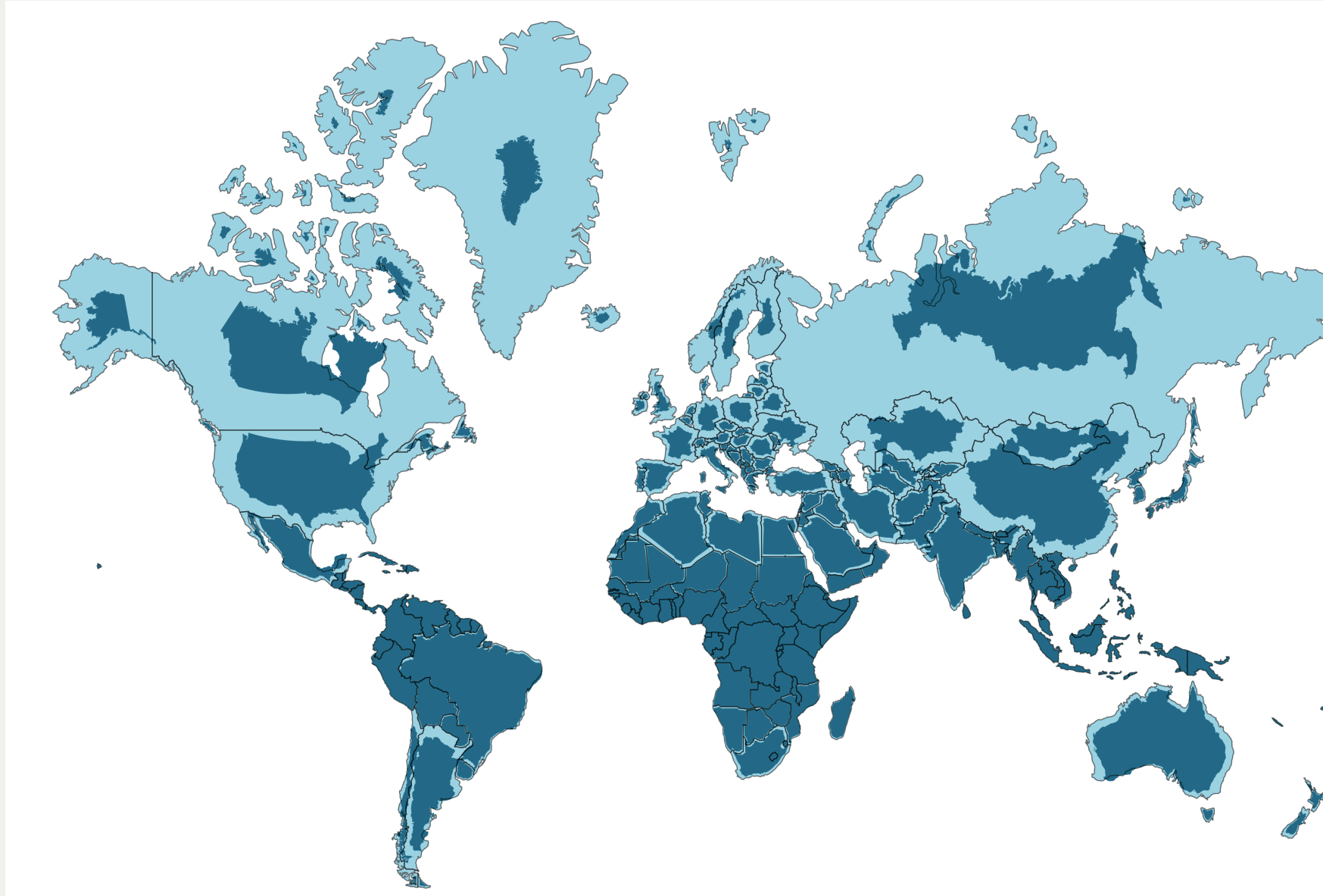
- **Mercator Projection:** Preserves shape but distorts area, especially near the poles. Commonly used for navigation.
- **Robinson Projection:** A compromise projection that attempts to balance distortions of area, shape, distance, and direction. Often used for world maps.
- **Albers Equal-Area Conic Projection:** Preserves area but distorts shape, distance, and direction.
 - Ideal for presenting particular regions of the world. Commonly used for maps of the United States.

There is no perfect projection! Pick the one that best suits your needs.

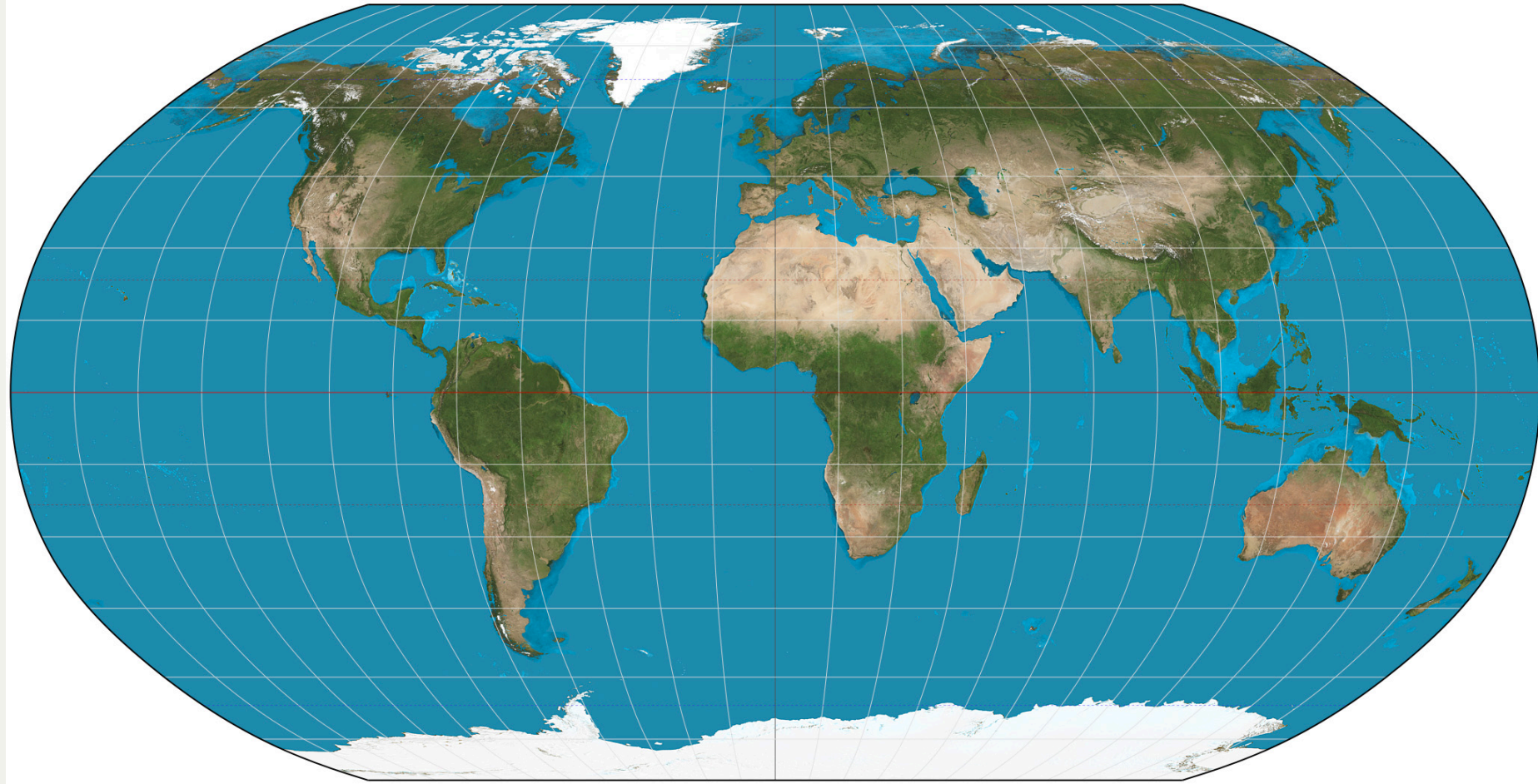
Mercator Projection



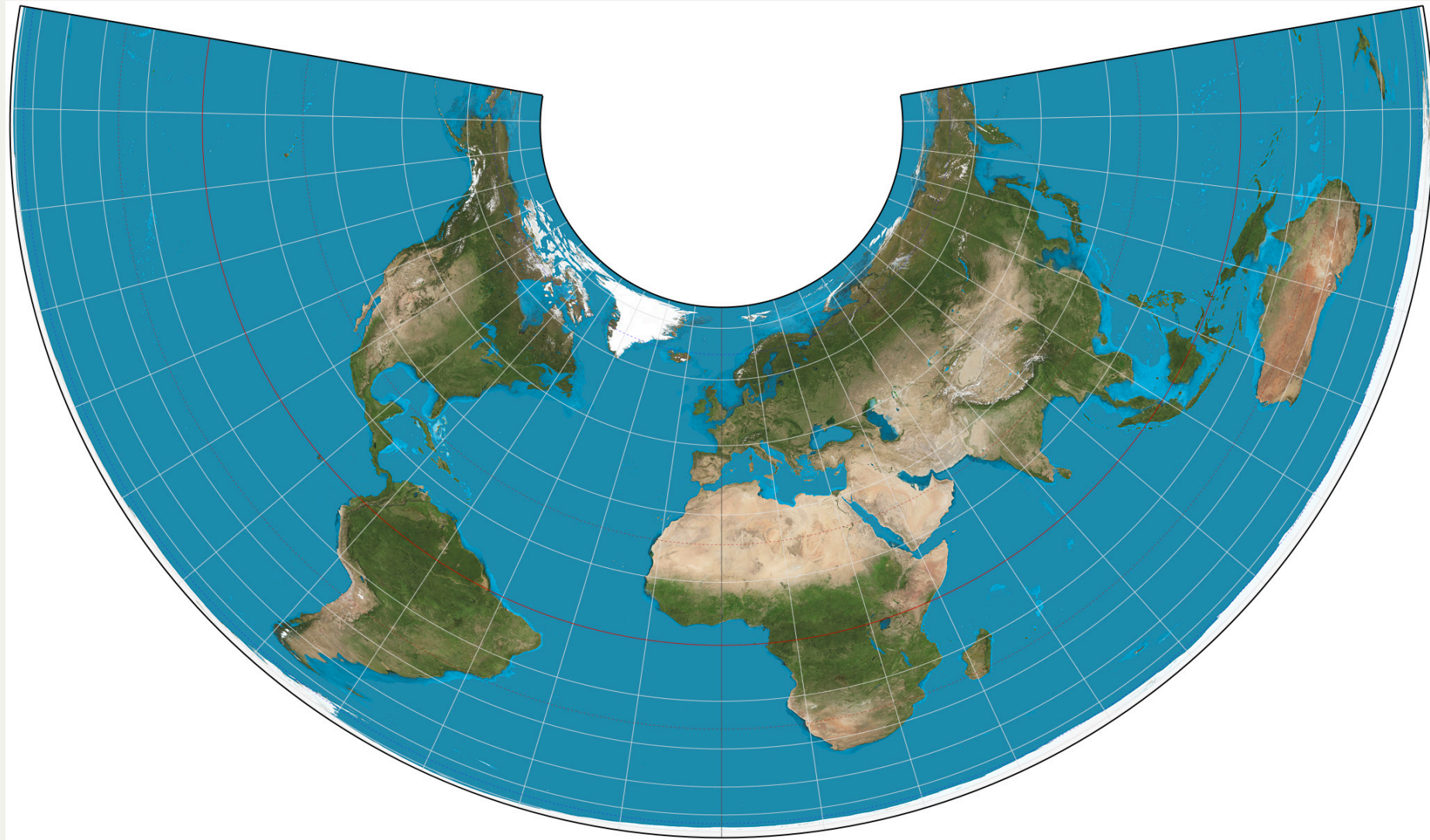
Mercator Projection Actual Size



Robinson Projection



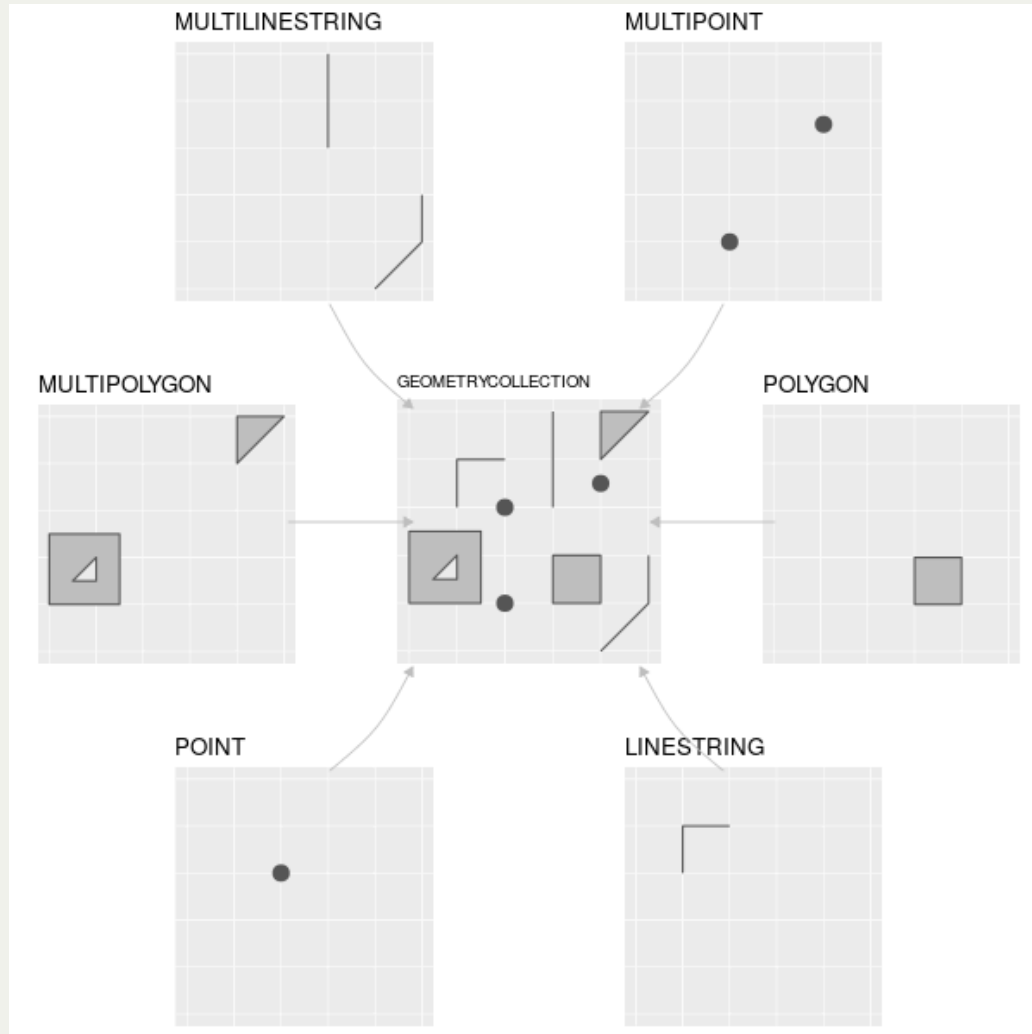
Albers Equal Area Conic Projection



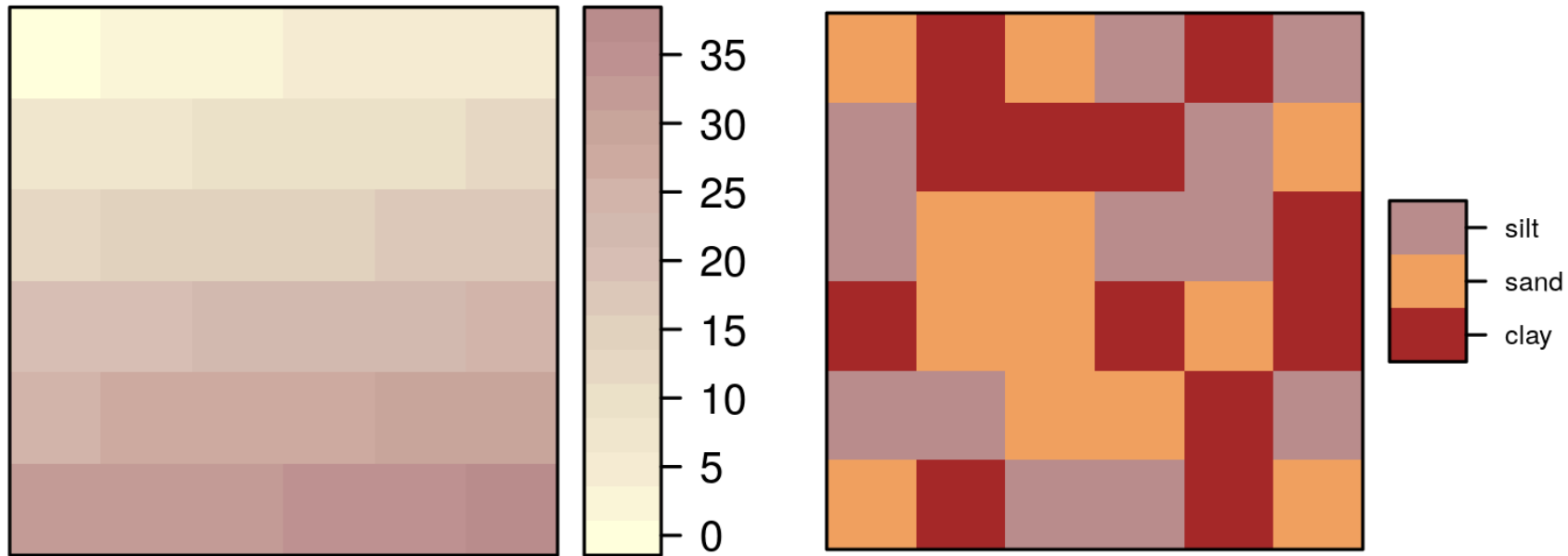
Geospatial Data Formats

- **Vector Data:** Represents geographic features as points, lines, and polygons. Common formats include:
 - Shapefiles (.shp)
 - GeoJSON (.geojson)
 - KML (.kml)
- **Raster Data:** Represents geographic features as a grid of pixels. Common formats include:
 - GeoTIFF (.tif)
 - NetCDF (.nc)

Vector Data



Raster Data



Mapping software

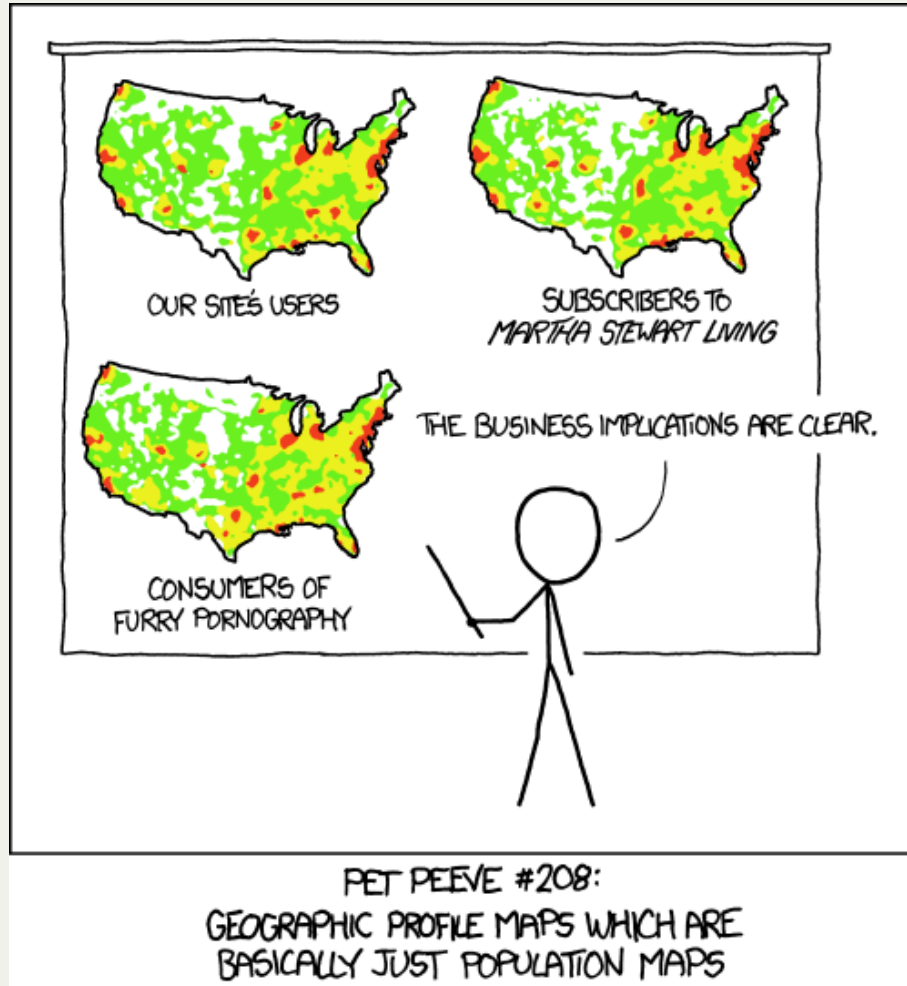
- **QGIS:** A free and open-source desktop GIS software that allows you to create, edit, visualize, analyze, and publish geospatial information.
- **ArcGIS:** A commercial GIS software suite developed by Esri that provides tools for working with geospatial data, including mapping, analysis, and data management.
- **R:** A programming language and environment for statistical computing and graphics, which has several packages (e.g. ggplot2, sf, leaflet) for creating maps and working with geospatial data.

FYI: For maps, R is the best! Free, open-source, replicable via code, and has a huge community of users.

Data Sources

- **National Oceanic and Atmospheric Administration (NOAA):**
 - An incredible source for data
 - Has everything from light at night to forest cover estimates
- **NYC Open Data/Data2Go**
 - Everything from transit stops to squirrel sightings
 - Neighborhood level Human Development Index!

Common Pitfall



- A shockingly common mistake you can't miss once you know how to look for it
- Can be addressed by adjusting for population